Project Report On

Ransomware Behaviour Detection Model

Submitted in partial fulfillment of the requirements for the award of the degree of

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in

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CERTIFICATE

This is to certify that the project report entitled ***"Ransomware Behaviour Detection Model"*** *is a bonafide record of the work done by* ***Dev Sebastian Joseph (U2203077)***, ***Joel John Thumpayil (U2203116)***, ***Golda John (U2203102)*** *and* ***Joes Jacob Paul (U2203117)*** *submitted to the Rajagiri School of Engineering & Technology (RSET) (Autonomous) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in "Computer Science and Engineering" during the academic year 2025-2026.*

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Abstract

This project presents a modular, real-time ransomware detection framework that integrates behavioral monitoring, sequence learning, and lightweight virtualization-based mitigation to protect host systems from evolving ransomware attacks. The system combines machine learning based anomaly detection with hypervisor level process introspection to provide early, explainable, and resilient defense against both known and zero-day ransomware variants. A dual stage detection pipeline leverages LightGBM for fast, feature based classification of filesystem and registry activity, while an LSTM based temporal model analyzes system call sequences to identify malicious encryption behavior patterns over time. A BitVisor based lightweight hypervisor operates as an inline security layer, enabling transparent process isolation and secure file backup without modifying the guest operating system. To adapt to dynamic ransomware behavior, the system incorporates continuous feedback-driven model refinement and intelligent feature pipelining, improving generalization across ransomware families. The decision engine fuses outputs from both classifiers to improve robustness and reduce false positives. When ransomware activity is detected, a rapid response module halts malicious encryption, restores affected files, and logs incidents for forensic analysis. Overall, this system demonstrates an efficient and scalable approach to proactive ransomware defense by combining machine learning, behavioral analysis, and hardware-assisted virtualization.

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Chapter 1

Introduction

This chapter lays the foundation for the Ransomware Behaviour Detection Model , outlining the core problem posed by the increasing sophistication and frequency of ransomware attacks in modern computing environments. It details the project's background, objectives, scope, and overall relevance. The subsequent sections define the key challenges and assumptions that guide this work.

1.1 Background

Ransomware has rapidly evolved into one of the most disruptive and financially damaging forms of cybercrime. It is a class of malware that encrypts a victim's files or locks access to critical system functions, demanding a ransom payment for restoration. Modern ransomware families like WannaCry, LockBit, and Ryuk have demonstrated the ability to spread rapidly across networks, evade traditional security tools, and cause large-scale data breaches and system downtime.

Conventional antivirus and intrusion detection systems rely heavily on signature-based detection, which compares programs against known malware patterns. However, with the rise of zero-day ransomware, polymorphic malware, and fileless attacks, signature-based defenses are no longer sufficient. These advanced ransomware variants modify their execution patterns, use obfuscation techniques, and exploit trusted system processes to remain undetected. Additionally, ransomware can begin file encryption within seconds of execution, leaving minimal time for defensive action.

To address these limitations, cybersecurity research has shifted toward behavior-based detection, which identifies ransomware by analyzing anomalous system behavior rather than static signatures. Machine learning models such as LightGBM and LSTM have shown

strong potential in detecting malicious activity based on system call sequences and file access patterns. Furthermore, integrating detection mechanisms with virtualization-based security layers like BitVisor enables early interception and safe mitigation actions, even before encryption is completed.

This project is motivated by the urgent need for a proactive and intelligent ransomware defense system capable of detecting both known and emerging ransomware strains in real time. By combining machine learning, behavioral analysis, and virtualization-based isolation, this system aims to deliver fast, explainable, and reliable protection against ransomware attacks.

1.2 Problem Definition

The central problem addressed by this project is the design and development of an intelligent ransomware detection and prevention system capable of identifying malicious activity at its earliest stages and preventing irreversible file encryption. Traditional signature-based antivirus solutions are largely ineffective against modern ransomware due to obfuscation, polymorphic behavior, and zero-day attacks. Consequently, the system must utilize behavioral analysis and machine learning techniques to detect both known and previously unseen ransomware variants in real time. Key challenges include achieving early detection before significant file or system damage occurs, accurately identifying zero-day threats that do not match known signatures, minimizing false positives to avoid disruption of legitimate system processes, and implementing a real-time response mechanism that halts ransomware activity and protects files without degrading system performance.

1.3 Scope and Motivation

Scope

The scope of this project encompasses the development of a hybrid ransomware detection and prevention framework that combines machine learning techniques with virtualization-based defense mechanisms. The system is designed to function at both the host and hypervisor levels, enabling proactive mitigation of ransomware attacks. Key functionalities include feature-based static detection using LightGBM to analyze system logs and file

behavior patterns for rapid identification of threats, behavioral dynamic detection through an LSTM model to monitor sequential system calls and uncover suspicious activity over time, virtualization-based defense via BitVisor to isolate malicious processes and protect critical files, and a secure response pipeline that facilitates rapid process termination, system recovery, and secure backup operations upon detection of ransomware.

Motivation

The motivation for this project arises from the increasing frequency, sophistication, and economic impact of ransomware attacks across industries. Modern ransomware is capable of evading traditional antivirus defenses through obfuscation and encryption, causing permanent data loss and business downtime within minutes, and spreading laterally across networks to compromise critical infrastructure. These threats highlight the urgent need for a proactive, intelligent defense mechanism that combines real-time behavioral analysis with secure system interventions to detect and prevent ransomware before significant damage occurs. This project aims to address these challenges by developing a hybrid detection model that is accurate, explainable, and practical for deployment in real-world cybersecurity environments.

1.4 Objectives

The key objectives of this project are to develop a comprehensive ransomware detection and prevention system that operates in real time. This includes creating a hybrid detection pipeline that leverages LightGBM for static feature-based analysis and LSTM for behavioral sequence monitoring. A BitVisor-based virtualization layer will be implemented to intercept and mitigate ransomware at the hypervisor level, ensuring early containment of malicious processes. The system will incorporate a secure response mechanism to halt ransomware execution and restore affected files promptly. Additionally, the project aims to achieve high detection accuracy while minimizing false positives through careful feature engineering and model optimization. Finally, detailed logging and explainable outputs will be provided for every detection event to facilitate forensic analysis and support further system improvement.

1.5 Challenges

The development and implementation of this system pose several technical challenges and risks. One major challenge is obtaining real, diverse, and balanced ransomware datasets that accurately represent real-world attack behaviors, as limited or biased data can hinder model performance. Another difficulty lies in ensuring model generalization so that the system can effectively detect zero-day ransomware variants that deviate from known attack patterns. Additionally, achieving real-time detection while minimizing performance overhead is essential to avoid slowing down legitimate system operations. Finally, integrating machine learning models with a hypervisor-based defense framework such as BitVisor introduces architectural complexity, requiring careful coordination between system components to ensure reliable and secure functionality.

1.6 Assumptions

This project proceeds based on several key assumptions. It assumes the availability of labeled ransomware datasets and benign system activity logs required for model training and evaluation. Sufficient computational resources, including GPU support for training machine learning models and hardware compatibility for BitVisor-based virtualization, are also presumed to be accessible. Additionally, it is assumed that the target deployment environment permits the installation and execution of hypervisor-level security modules without violating system constraints or administrative policies. Finally, the project assumes that user or system feedback will be available to support iterative improvement of detection accuracy and overall model performance.

1.7 Societal / Industrial Relevance

This project holds significant relevance for both industry and society due to the increasing prevalence and sophistication of ransomware attacks. By providing proactive protection mechanisms, it strengthens the cybersecurity posture of individuals, organizations, and government systems. The system's ability to detect and prevent ransomware in real time can greatly reduce financial losses associated with downtime and data recovery, offering substantial economic benefits. Furthermore, this work advances research in cybersecurity

by integrating machine learning techniques with virtualization-based defense mechanisms to enhance malware detection accuracy and resilience. Owing to its modular architecture and lightweight implementation, the proposed system is scalable and suitable for deployment across a wide range of enterprise environments, making it practical for real-world industrial applications.

1.8 Organization of the Report

The report is organized into four chapters. Chapter 1 provides an introduction to the project and its objectives. Chapter 2 presents a review of related research. Chapter 3 explains the system design and methodology. Chapter 4 concludes the report with a summary and future scope.

This chapter introduced the ransomware detection project, highlighted the limitations of traditional malware defenses, defined the problem of early ransomware detection, and outlined the project's objectives, challenges, assumptions, and real-world relevance.

Chapter 2

Literature Survey

This chapter presents an overview of recent research on ransomware detection, highlighting the evolution of hybrid machine learning models, deep learning-based anomaly detection, deception-based defense mechanisms, and forensic classification methods. The reviewed works emphasize scalable, real-time, and lightweight approaches for identifying ransomware activity effectively across diverse environments.

2.1 Review of Key Research Works

2.1.1 Hybrid Ransomware Detection Mechanisms [1]

Hybrid ransomware detection mechanisms aim to overcome the shortcomings of traditional static and dynamic detection methods by combining multiple analytical layers into a unified framework. These approaches integrate static signature analysis, dynamic behavioral monitoring, and artificial intelligence–based feature learning to detect both known and unknown ransomware variants. By leveraging machine learning and deep learning models such as Autoencoders, CNN–RNN hybrids, and LightGBM, these systems provide a more comprehensive understanding of malicious behaviors. The hybrid model enhances detection accuracy and reduces false positives through multi-layered feature correlation, achieving real-time performance suitable for endpoint deployment. Although this approach offers significant advantages in robustness and adaptability, it also introduces challenges related to implementation complexity and computational overhead, particularly when monitoring large-scale or high-traffic environments.

2.1.2 LSTM Autoencoder-based Ransomware Detection [2]

LSTM Autoencoder-based ransomware detection models are designed to identify unknown or zero-day ransomware through deep anomaly detection. These models use Long Short-Term Memory (LSTM) Autoencoders to learn normal system behaviors by analyzing Input/Output Request Packet (IRP) sequences over time. Any deviation from the learned behavior pattern is classified as anomalous and flagged as potential ransomware activity. The LSTM Autoencoder's ability to capture temporal dependencies in system processes allows for early identification of malicious activity before significant encryption damage occurs. Experimental evaluations demonstrated a True Positive Rate (TPR) of 97.8% and a False Positive Rate (FPR) of 2.1%, highlighting its high detection precision and reliability. This deep learning approach provides a robust framework for early detection of previously unseen ransomware, but it requires large volumes of quality data and considerable computational power for training, which can limit its real-time applicability on resource-constrained systems.

2.1.3 Deception-based Defenses for Ransomware [3]

Deception-based defenses introduce a proactive layer of protection against ransomware by luring malicious software into interacting with decoy files instead of valuable user data. These mechanisms employ honeyfiles, fake directories, and ransomware-aware filesystems to detect and divert encryption activities before they cause harm. When ransomware engages with these decoys, the system immediately recognizes the activity and initiates defensive actions, such as isolating the affected process or halting system access. This method significantly reduces data loss and allows for the rapid containment of attacks. Deception-based defenses are lightweight, practical, and can be seamlessly integrated into existing endpoint protection systems. Their main strength lies in early attack interception and minimal computational cost. However, their effectiveness diminishes against ransomware variants that selectively target files or avoid interacting with deceptive resources, requiring continuous adaptation to attacker tactics.

2.1.4 Forensic Analysis and Classification of Ransomware Families [4]

Forensic analysis and classification approaches to ransomware focus on understanding and categorizing ransomware families through behavioral and statistical analysis. These studies utilize dynamic execution tracing, system call monitoring, and feature extraction techniques to identify distinguishing behavioral signatures of various ransomware strains. Machine learning classifiers are trained on these features to achieve accurate and resilient classification, even in the presence of code obfuscation or packing techniques. This forensic approach not only improves detection accuracy but also enhances incident response and malware attribution. By providing insights into the operational behavior and evolution of ransomware families, forensic classification frameworks aid in proactive defense and recovery strategies. Despite their analytical power, such models depend heavily on runtime data and controlled execution environments, which may limit their efficiency when applied to large-scale or live systems.

2.1.5 Ransomware Detection using Deep Learning–based Unsupervised Feature Extraction and Pareto Ensemble Learning [5]

Recent advancements in deep learning and ensemble optimization have introduced models that combine unsupervised feature extraction with Pareto ensemble learning for ransomware detection. In these systems, Autoencoders are used to extract deep latent features from input data without the need for labeled samples. Pareto optimization techniques are then applied to balance multiple objectives, including detection accuracy, false positive rate, and computational cost. This approach ensures that the detection framework remains effective across diverse ransomware behaviors and evolving attack patterns. The ensemble structure enhances robustness and generalization by integrating the strengths of multiple learners. The resulting system achieves a well-balanced trade-off between sensitivity and specificity, making it suitable for real-world, large-scale network environments. However, the framework's complexity and high resource requirements for training and optimization remain a major limitation, necessitating further refinement for lightweight, real-time deployment.

2.2 Comparative Analysis

Paper	Approach	Model Type	Strength	Limitation
Hybrid Ransomware Detection Mechanisms [1]	Static + Dynamic + AI	Hybrid ML / DL	Real-time and accurate detection	High computational cost
LSTM Autoencoder-based Detection [2]	Sequence anomaly learning	Deep Learning (LSTM)	Detects unknown variants early	Requires large datasets
Deception-based Defenses [3]	Bait and decoy mechanisms	Deception Model	Lightweight and practical	Limited coverage of stealth attacks
Forensic Analysis and Classification [4]	Dynamic monitoring	ML Classifiers	Accurate and resilient to obfuscation	High runtime dependency
DL-based Unsupervised Feature Extraction with Pareto Ensemble [5]	Feature extraction + Pareto optimization	Autoencoder + Ensemble	Balanced accuracy and scalability	Training resource intensive

Table 2.1: Comparison of Recent Ransomware Detection and Defense Approaches

2.3 Summary and Gaps Identified

The reviewed works collectively demonstrate that hybrid detection systems and deep learning based models have significantly advanced ransomware detection capabilities. LSTM Autoencoders and Pareto ensemble learning frameworks have enabled the early and balanced detection of ransomware through temporal and feature-based anomaly analysis. Deception-based defenses add an additional layer of resilience by diverting ransomware from critical assets and providing valuable detection signals. However, current approaches face challenges such as high computational complexity, lack of standardized evaluation datasets, and limited adaptability across heterogeneous platforms. Future research should prioritize lightweight hybrid detection frameworks that combine behavioral analytics, deception mechanisms, and adaptive machine learning to provide scalable and efficient protection against evolving ransomware threats.

Chapter 3

System Design

3.1 Overall System Architecture

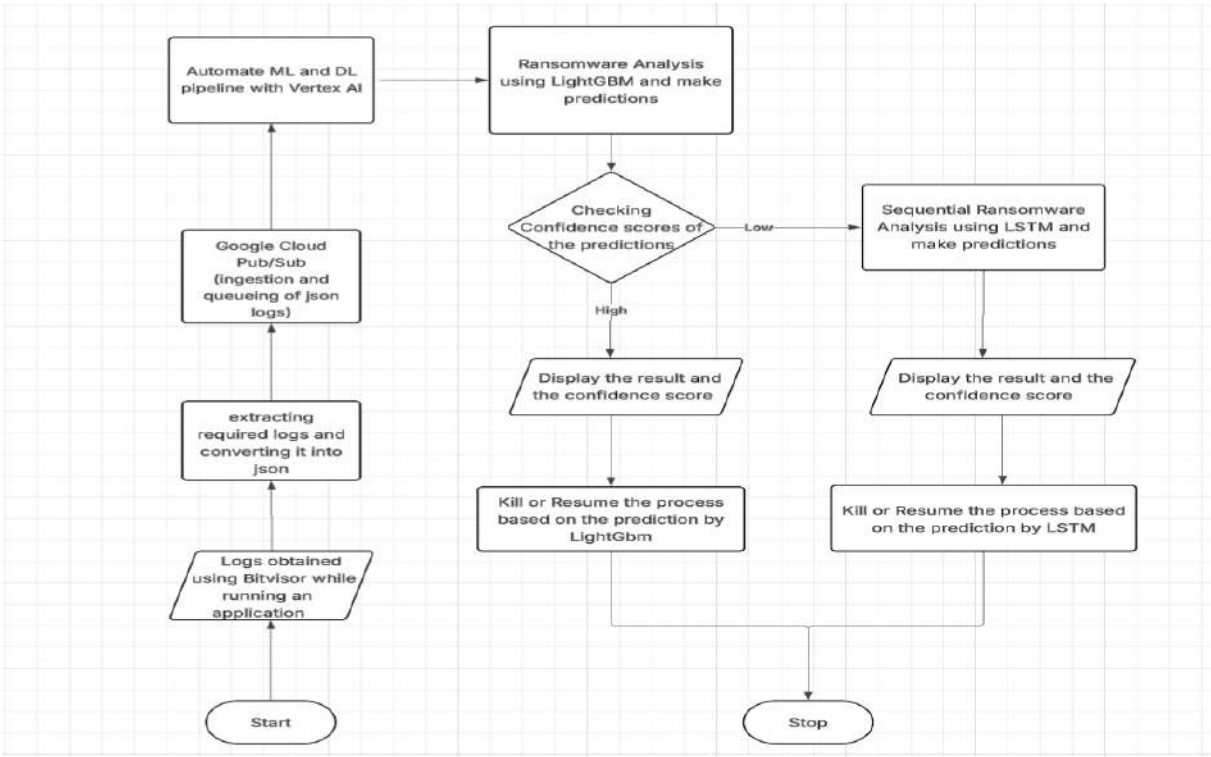

Figure 3.1: System Architecture of the Ransomware Detection Model

3.1.1 Input Module

From Figure 3.1, The Input Module is responsible for gathering system activity data, including system logs, file access events, registry modifications, and process information from the host. It then preprocesses this data by cleaning, normalizing, and structuring it to make it suitable for model input. Key features relevant for ransomware detection, such as file write patterns, process behavior, and registry changes, are extracted and

transformed into numerical representations through vectorization, making them compatible with LightGBM and LSTM models. Finally, dense feature embeddings are generated to capture sequential and semantic patterns in system activity, enabling accurate and timely detection of ransomware behavior.

3.1.2 Detection Module

The Detection Module combines multiple approaches to identify ransomware activity effectively. It uses a static feature classifier based on LightGBM to analyze extracted file and system features, while a behavioral sequence analyzer employing LSTM monitors system call sequences to detect suspicious temporal patterns. Outputs from both models are fused through a candidate fusion process to improve overall detection accuracy and minimize false positives. Additionally, a threat database is maintained, containing known ransomware signatures, behavioral patterns, and family information, which serves as a reference for both detection and verification purposes.

3.1.3 Session Context Module

The Session Context Module manages and stores system metadata, including system IDs, timestamps, process IDs, and prior alerts, to ensure continuity in monitoring. It tracks ongoing system activity, retrieves relevant historical patterns, and maintains temporal coherence, providing contextual grounding for ransomware detection by considering previous alerts and suspicious behaviors. This module also supports multi-step decision-making, enabling the system to accurately identify threats by analyzing activity trends over time and correlating them with past incidents.

3.1.4 Adaptive Learning Module

The Adaptive Learning Module continuously improves the ransomware detection system by refining the LightGBM and LSTM models using newly collected system logs and confirmed threat labels. It performs hyperparameter optimization to balance detection sensitivity and minimize false positives, while dynamically updating known ransomware behaviors and detection rules based on observed incidents. By learning from emerging ransomware patterns, this module enhances the system's adaptability and overall detection accuracy, ensuring it remains effective against both known and novel threats.

3.1.5 Feedback Loop

The Feedback Loop monitors and evaluates the performance of the ransomware detection system by verifying alerts to determine whether they are true positives, false positives, or missed threats. It tracks key performance metrics, including detection accuracy, precision, recall, F1 score, latency, and the success of mitigation actions. All outcomes from detection and response activities are logged in a feedback store, enabling continuous refinement of the models and improvement of response strategies. This iterative feedback process ensures that the system adapts over time, enhancing detection accuracy and operational effectiveness against evolving ransomware threats.

3.1.6 Response and Mitigation Module

The Response and Mitigation Module manages ransomware threats by using BitVisor to isolate suspicious processes and prevent further file encryption. It executes recovery actions by restoring affected files from secure backups or snapshots whenever possible, and generates actionable alerts for administrators with detailed information about the detected threats. This module ensures timely and effective mitigation of ransomware attacks while minimizing disruption to normal system operations, providing a robust defense layer that complements detection and adaptive learning components.

3.2 Design Diagrams

3.2.1 Detailed System Architecture

The detailed system architecture of the proposed ransomware detection system is illustrated in Figure 3.2. The architecture follows a layered pipeline that enables continuous monitoring, feature extraction, machine learning based detection, and automated response to ransomware threats.

The process begins with system activity monitoring through BitVisor, which captures low-level disk and memory operations from running applications. These logs are then processed and converted into structured features that are suitable for machine learning analysis. The extracted features are analyzed by the ransomware detection engine, which integrates LightGBM for fast statistical classification and an LSTM network for sequential

behavioral analysis. Based on the detection results, the system generates alerts and initiates mitigation actions such as isolating suspicious processes and preventing further file encryption.

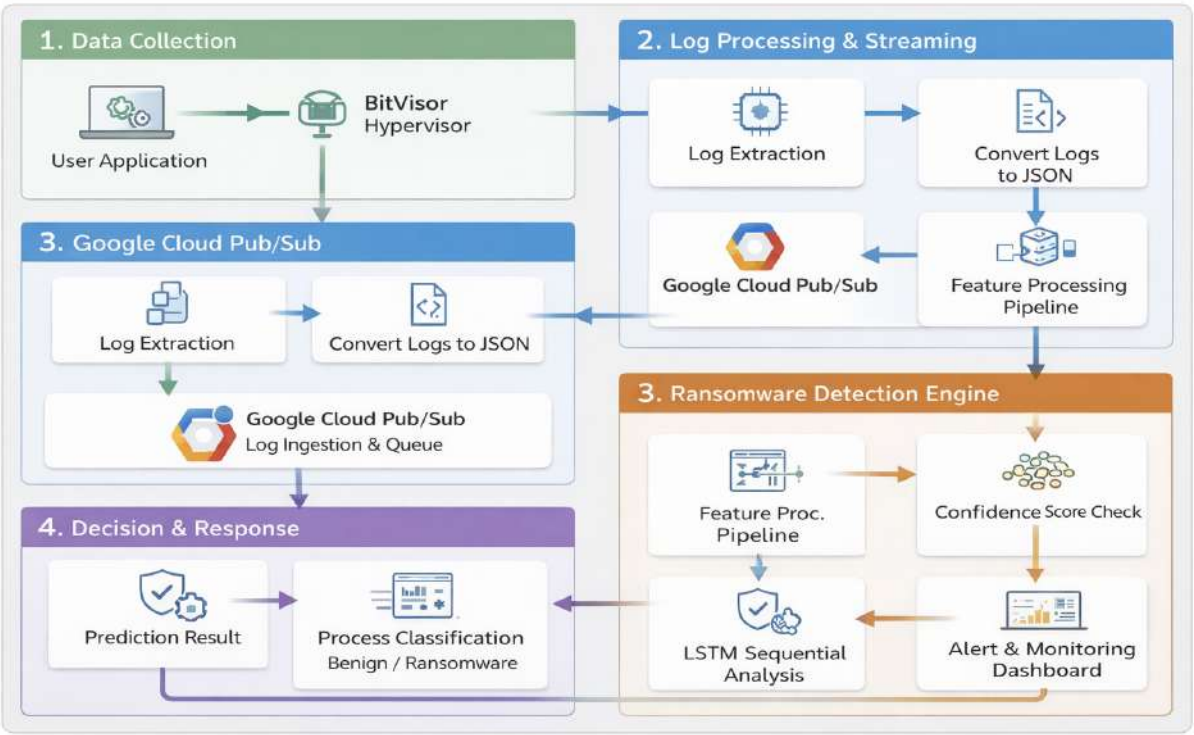

Figure 3.2: Detailed System Architecture of the Ransomware Detection System

3.2.2 Sequence Diagram

This section illustrates the order of interactions among system modules during ransomware detection and mitigation shown in Figure 3.2. The process begins with System Monitoring, which collects real-time system activity such as file access, process execution, and registry changes. The Input Module then preprocesses this data and extracts relevant features for analysis. The Session Context Module enriches the input with historical system activity and prior alerts, providing temporal and contextual grounding. The Detection Module applies LightGBM and LSTM models to identify potential ransomware activity, after which the Response and Mitigation Module isolates suspicious processes using BitVisor, halts encryption, and restores affected files. The Feedback Loop records detection outcomes, verifies alerts, and tracks performance metrics, while the Adaptive Learning Module updates model parameters and threat patterns based on this feedback to improve

future detection accuracy. Throughout the process, the Threat Database supplies known ransomware signatures and behaviors and receives updates from new detections, ensuring continuous learning and system improvement.

Cycle Summary: System Activity → Feature Extraction → Detection → Mitigation → Feedback → Model Update → Enhanced Detection.

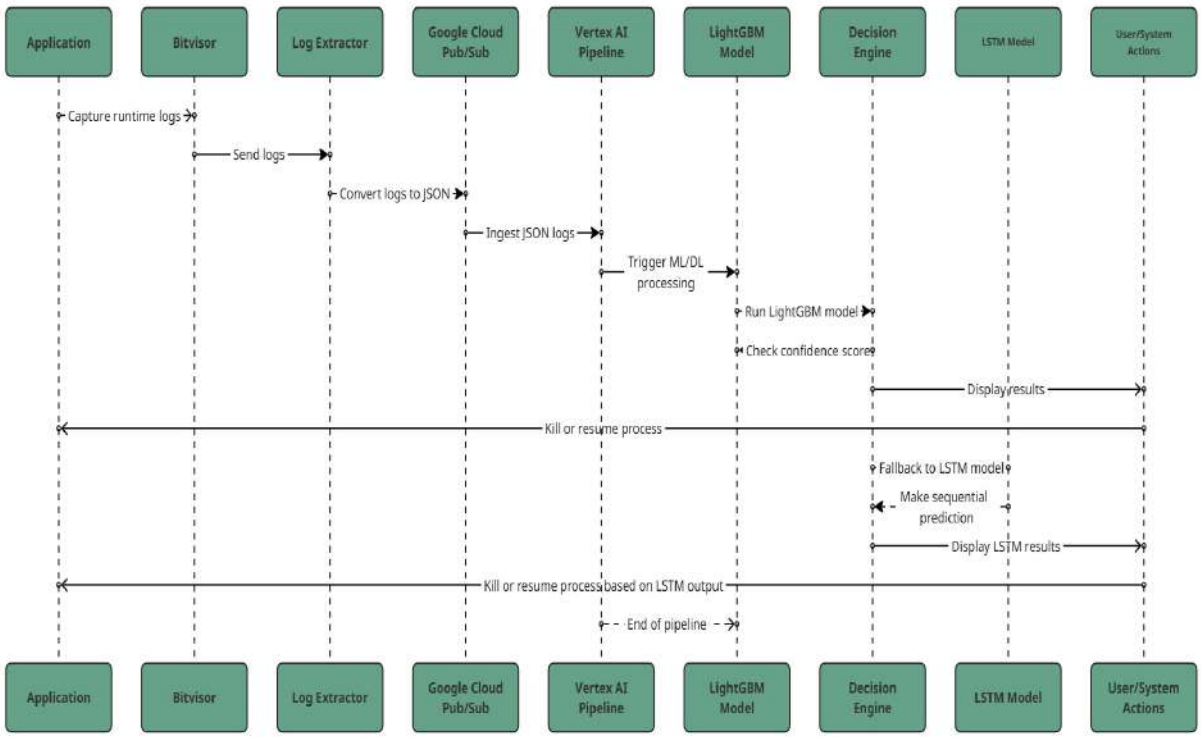

Figure 3.3: Sequence Diagram of the Ransomware Detection Model

3.3 Algorithms Used

This section details the core algorithms powering the ransomware detection system. The project leverages a combination of LightGBM for static feature classification and LSTM for behavioral sequence analysis, integrated with a candidate fusion mechanism to enhance detection accuracy.

3.3.1 LightGBM for Static Feature Classification

LightGBM is a gradient boosting framework that builds decision trees sequentially, optimizing for a differentiable loss function.

Working: Given a training dataset $\{(x_i, y_i)\}_{i=1}^{N}$ with features $x_i \in \mathbb{R}^d$ and labels $y_i \in \{0, 1\}$ (benign or ransomware), LightGBM minimizes the loss function:

$$L = \sum_{i=1}^{N} \ell(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k)$$

where ℓ is a differentiable loss (e.g., binary cross-entropy), f_k are the decision trees, and $\Omega(f_k)$ is the regularization term for complexity control.

Algorithm 1 LightGBM Training for Ransomware Detection

Require: Training set $\{(x_i, y_i)\}_{i=1}^{N}$, number of trees K

Ensure: Trained LightGBM model

```
Initialize predictions: ŷ_i = 0 for all i
for k = 1 to K do
    Compute gradients g_i = ∂ℓ(y_i, ŷ_i) / ∂ŷ_i
    Fit a decision tree f_k to g_i
    Update predictions: ŷ_i ← ŷ_i + η f_k(x_i)
end for
Result: Trained LightGBM model
```

:

3.3.2 LSTM for Behavioral Sequence Analysis

Core Idea: Long Short-Term Memory (LSTM) networks capture temporal dependencies in sequential data, such as system call sequences or file access events.

Application in the Project: LSTM analyzes sequences of system events to detect abnormal behavioral patterns indicative of ransomware.

Working: At each timestep t, given input x_t, previous hidden state h_{t-1}, and previous cell state C_{t-1}:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad \text{(forget gate)}$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad \text{(input gate)}$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad \text{(cell candidate)}$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad \text{(cell state update)}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad \text{(output gate)}$$

$$h_t = o_t \odot \tanh(C_t) \quad \text{(hidden state)}$$

Algorithm 2 LSTM-Based Behavioral Detection

Require: Sequence of system events $X = [x_1, x_2, ..., x_T]$

Ensure: Probability of ransomware $p \in [0, 1]$

```
Initialize h_0 = 0, C_0 = 0
for t = 1 to T do
    Compute f_t, i_t, C̃_t, C_t, o_t, h_t using above equations
end for
Compute output: p = σ(W_out h_T + b_out)
Result: p
```

:

3.3.3 Candidate Fusion

Core Idea: Candidate fusion combines predictions from LightGBM and LSTM models.

Working: Given LightGBM output p_{LGB} and LSTM output p_{LSTM}, the fused probability is computed as:

$$p_{fused} = \alpha \cdot p_{LGB} + (1 - \alpha) \cdot p_{LSTM}, \quad \alpha \in [0, 1]$$

The final decision is:

$$\hat{y} = \begin{cases} 1 & \text{if } p_{fused} \geq \tau \\ 0 & \text{otherwise} \end{cases}$$

Algorithm 3 Candidate Fusion for Ransomware Detection

Require: LightGBM probability p_{LGB}, LSTM probability p_{LSTM}, fusion weight α, threshold τ

Ensure: Final detection $\hat{y}$

```
Compute fused probability: p_fused = α · p_LGB + (1 − α) · p_LSTM
if p_fused ≥ τ then
    ŷ ← 1 (ransomware)
else
    ŷ ← 0 (benign)
end if
Result: ŷ
```

3.3.4 Performance Metrics

Metrics Used:

- Accuracy: $\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$
- Precision: $\text{Precision} = \frac{TP}{TP+FP}$
- Recall: $\text{Recall} = \frac{TP}{TP+FN}$
- F1 Score: $\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$

3.4 Requirements

3.4.1 Software Requirements

Software/Frameworks

- **Core Framework:**
  - Custom ransomware detection modules implementing the LightGBM and LSTM models

- **Detection Tools:**
  - LightGBM – Static feature-based ransomware detection
  - LSTM – Behavioral sequence analysis for dynamic detection
  - BitVisor – Hypervisor-level monitoring and process isolation for mitigation
- **Development Tools:**
  - VS Code / Other IDE – Development and testing
  - Git + GitHub – Version control

3.4.2 Hardware Requirements

- **Minimum:**
  - 4-Core CPU
  - 16 GB RAM
  - GPU with 4–6 GB VRAM for LSTM training
  - 50 GB Storage
  - BitVisor-compatible hardware for virtualization
- **Recommended:**
  - 8-Core CPU
  - $\geq$ 32 GB RAM
  - GPU with $\geq$ 12 GB VRAM

3.5 Key Deliverables

The key deliverable is a modular ransomware detection and mitigation system that detects ransomware using LightGBM for static feature analysis and LSTM for behavioral sequence monitoring, mitigates threats in real time via BitVisor by 000

malicious processes and preventing file encryption, maintains system context by tracking prior alerts, system activity and process behavior for accurate detection.

3.6 Project Timeline and Work Division

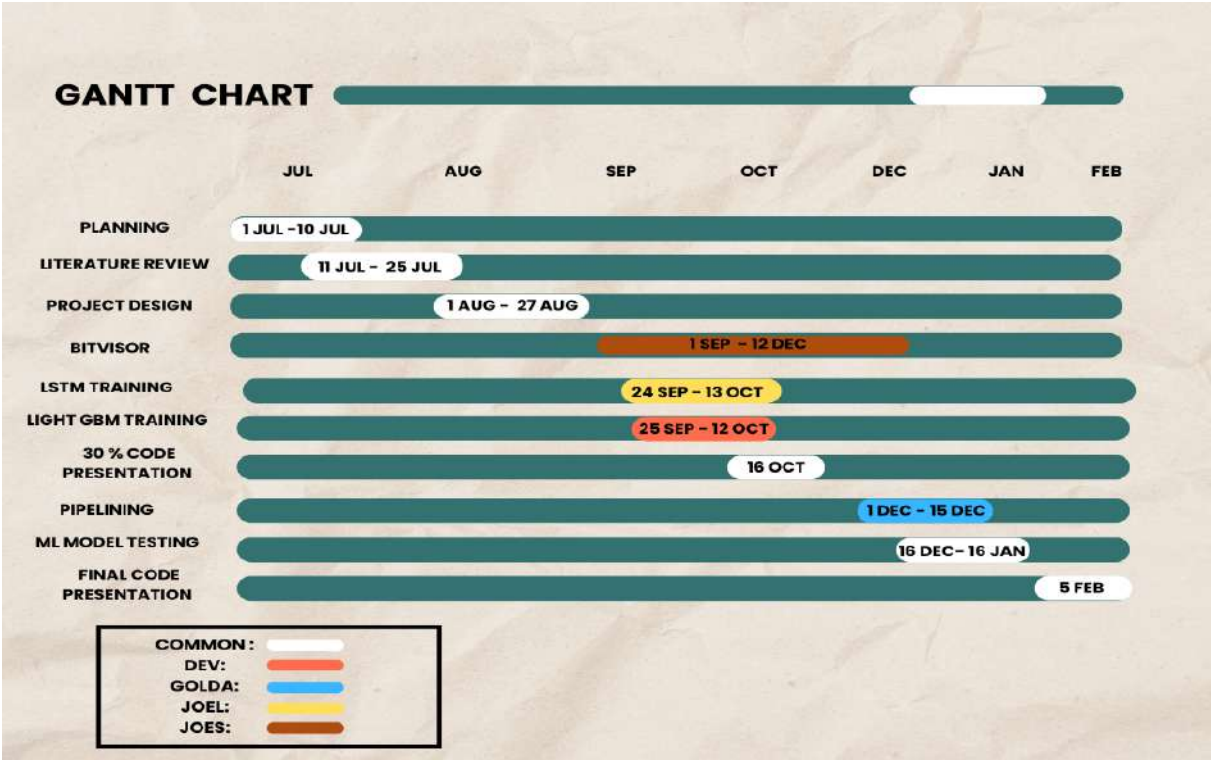

Figure 3.4: Gantt Chart of Work Division

From Figure 3.3, the Gantt Chart outlines the project timeline from July to February, showing tasks, durations and team responsibilities. Planning, literature review and project design occur from July to August, followed by training in late September and early October. Core development, including ML model testing, runs mainly from September to December. Presentation preparation happens in December, and final coding continues until January, with the final presentation scheduled for February. Most tasks are common to all members, while specific training and presentation roles are individual.

3.7 Summary of Chapter

The chapter outlines hardware requirements, system modules including LightGBM, LSTM, and BitVisor, and the key deliverables of the project. Overall, this chapter provides a comprehensive blueprint for implementing an effective, real-time ransomware defense system, ensuring clarity in design, functionality, and resource planning.

Chapter 4

Results and Discussions

This chapter presents the experimental evaluation and results of the proposed ransomware behaviour detection system. The results demonstrate the effectiveness of the hybrid detection approach that combines LightGBM for fast feature-based classification and LSTM for deep behavioural sequence analysis.

The system was evaluated using datasets generated from BitVisor-based system activity logs, which capture low-level disk and memory operations. These logs were processed through the preprocessing pipeline and used to train and test the machine learning models.

4.1 Feature Engineering and Dataset Preparation

The dataset used for model training was generated from system activity logs collected through the modified BitVisor environment. These logs contain detailed information about disk operations, memory accesses, and system events that occur during application execution.

To transform the raw logs into meaningful inputs for the machine learning models, multiple behavioural features were extracted and aggregated.

4.1.1 LightGBM Feature Set

The LightGBM model relies on statistical features extracted from disk write operations and entropy patterns of written data.

The aggregated features include:

- **count**: Total number of write operations
- **size_mean**: Mean size of write operations
- **size_std**: Standard deviation of write sizes

- **entropy_mean**: Mean entropy of written data
- **entropy_std**: Entropy variability
- **entropy_median**: Median entropy value
- **zero_entropy_frac**: Fraction of zero-entropy writes
- **high_entropy_frac**: Fraction of high-entropy writes
- **unique_lba_count**: Number of unique logical block addresses accessed
- **spike_entropy_count**: Sudden entropy transitions in data

These features help distinguish ransomware activity because encryption typically produces high entropy data and repetitive disk write patterns.

4.1.2 Memory and Disk Behaviour Features

Additional behavioural indicators were derived from memory and disk activity logs.

Memory features

- mem_write_count
- mem_avg_entropy
- mem_std_entropy
- mem_total_bytes
- mem_gpa_variance

Disk features

- disk_write_count
- disk_avg_entropy
- disk_total_bytes
- disk_lba_variance

4.2 LightGBM Model Performance

The LightGBM model acts as the first-stage classifier in the detection pipeline. It performs fast classification of system activity into benign or suspicious behaviour.

4.2.1 Training Performance

During the training phase, the model achieved near-perfect classification results.

- AUC Score: 1.000
- Average Precision: 1.000
- Accuracy: 100%

$$\begin{bmatrix} 50 & 0 \\ 0 & 70 \end{bmatrix}$$

4.2.2 Testing Performance

When evaluated on the testing dataset, the model maintained strong performance.

- AUC Score: 0.9517
- Average Precision: 0.9744
- Accuracy: 87.67%

4.3 LSTM Behavioural Sequence Analysis

To capture complex behavioural patterns that occur over time, an LSTM-based deep learning model was implemented as the second stage of the detection pipeline.

Sequential datasets were generated from the system logs for LSTM training.

- X_sequences.npy → shape (150,877, 50, 10)
- y_labels.npy → shape (150,877)
- run_map.npy → mapping of each sequence to source run

4.3.1 LSTM Model Performance

- Accuracy: 99%
- Precision: 0.98
- Recall: 0.99
- F1-score: 0.99

$$\begin{bmatrix} 6834 & 30 \\ 146 & 18824 \end{bmatrix}$$

4.4 Real-world Execution Results

The system was evaluated on real execution traces of both ransomware and benign applications.

4.4.1 Ransomware Detection Example

For a DarkSide ransomware execution sample:

- Total sequences analyzed: 1373
- Ransomware sequences detected: 1356
- Benign sequences detected: 17
- Ransomware probability: 98.76%

The system produced a final classification:

Final Decision: Ransomware

```
==============================
📊 DARKSIDE SEQUENCE RESULTS
==============================
Total sequences       : 1373
Ransomware sequences  : 1356
Benign sequences      : 17
Ransomware %          : 98.76%
Benign %              : 1.24%

==============================
🧠 DARKSIDE RUN DECISION
==============================
Average probability : 0.9994
FINAL DECISION      : RANSOMWARE
```

Figure 4.1: Detection output for DarkSide ransomware execution showing sequence classification results.

4.4.2 Benign Application Example

For a benign Firefox execution run:

- Total sequences analyzed: 987
- Benign sequences detected: 970
- Ransomware sequences detected: 17
- Benign probability: 98.28%

The system produced the final classification:

Final Decision: Benign

```
==============================
📊 FIREFOX SEQUENCE RESULTS
==============================
Total sequences       : 987
Ransomware sequences  : 17
Benign sequences      : 970
Ransomware %          : 1.72%
Benign %              : 98.28%

==============================
🧠 FIREFOX RUN DECISION
==============================
Average probability : 0.0426
FINAL DECISION      : BENIGN
```

Figure 4.2: Detection output for benign Firefox execution showing correct benign classification.

4.5 System Dashboard and Monitoring Results

The system includes a monitoring dashboard that displays detection results in real time.

- System status indicator
- Number of processes monitored
- Number of threats detected and blocked
- Detection logs with LightGBM and LSTM scores
- Time-stamped alerts for suspicious activity

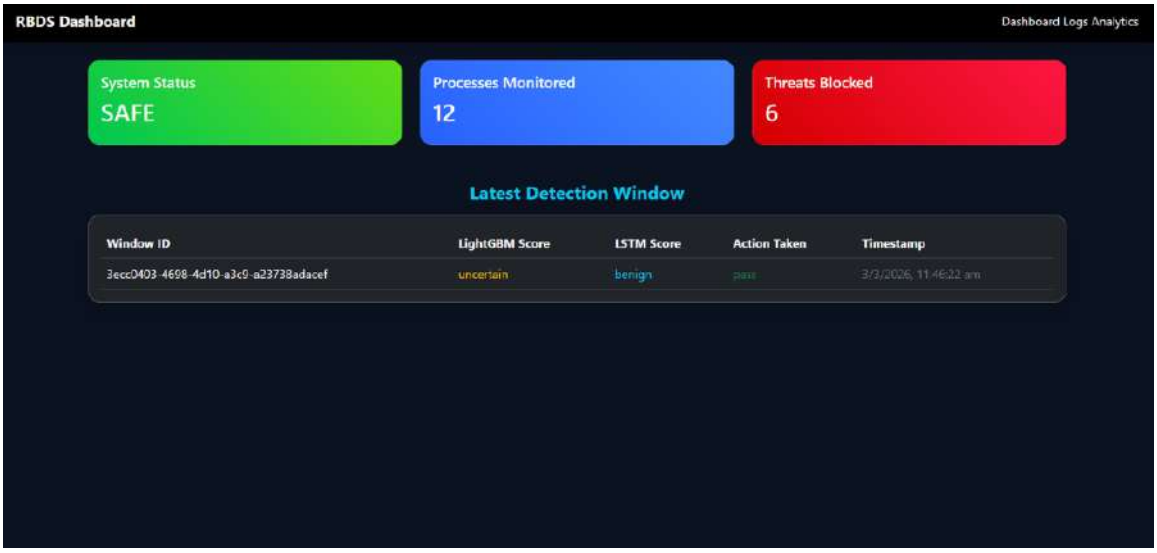

Figure 4.3: RBDS monitoring dashboard showing system status, processes monitored, and detected threats.

The analytics dashboard also visualizes the detection logs for monitored system activity.

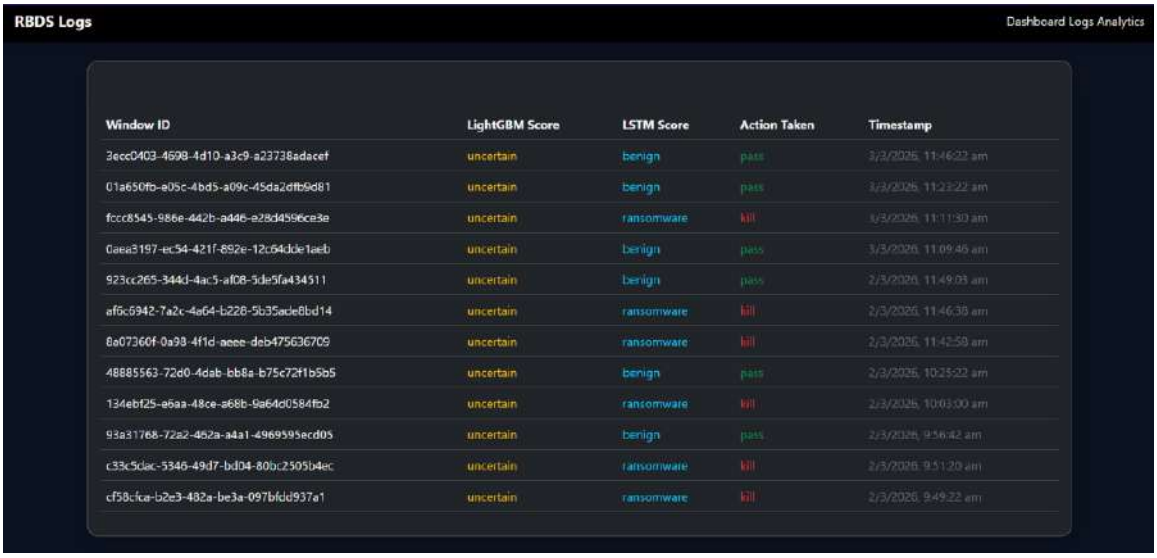

Figure 4.4: RBDS detection log table displaying LightGBM and LSTM scores, actions taken, and timestamps.

4.6 Discussion

The experimental results demonstrate that combining machine learning with behavioural analysis significantly improves ransomware detection performance.

LightGBM provides fast initial classification using statistical features extracted from

system logs. However, ransomware behaviour often evolves over time and cannot always be detected through single-event analysis.

The LSTM model complements LightGBM by analyzing sequential system behaviour and detecting temporal patterns associated with encryption activity.

The hybrid architecture therefore provides multiple advantages:

- Early detection before file encryption is completed
- Reduced false positives compared to static detection methods
- Ability to detect unknown ransomware variants
- Real-time monitoring capability

4.7 Summary of Chapter

This chapter presented the results and evaluation of the proposed ransomware behaviour detection system. The experimental results show that the hybrid LightGBM–LSTM detection pipeline achieves high detection accuracy and strong generalization capability.

Chapter 5

Conclusions & Future Scope

Ransomware attacks have become one of the most serious cybersecurity threats, causing significant financial losses and operational disruptions across organizations worldwide. Traditional signature-based detection methods are often unable to detect newly emerging or polymorphic ransomware variants. Therefore, this project focused on developing a behavior-based ransomware detection system capable of identifying malicious activity at an early stage before large-scale file encryption occurs.

The proposed system integrates machine learning techniques with low-level system monitoring to provide an effective ransomware detection framework. A hybrid detection pipeline combining LightGBM and LSTM models was implemented to analyze both statistical features and sequential behavioral patterns extracted from system logs. The LightGBM model enables fast classification of suspicious activity using aggregated disk and memory features, while the LSTM model captures temporal dependencies in system event sequences to detect hidden ransomware behavior patterns.

Experimental results demonstrate that the proposed system achieves high detection accuracy and effectively distinguishes between benign applications and ransomware executions. The LightGBM classifier provides rapid initial detection, while the LSTM model performs deeper behavioral analysis to improve detection reliability. The system was also evaluated using real execution traces, where ransomware samples such as DarkSide were correctly identified while legitimate applications such as Firefox were classified as benign. These results indicate that the hybrid detection approach can provide early and reliable detection of ransomware activity.

Another key contribution of this work is the integration of BitVisor-based monitoring, which enables the collection of low-level system activity without interfering with the guest operating system. This virtualization-layer monitoring allows the system to capture important behavioral indicators such as disk writes, memory access patterns, and entropy

changes, which are critical for detecting ransomware behavior.

Overall, the proposed ransomware behavior detection model demonstrates that combining behavioral monitoring, machine learning models, and virtualization-based security mechanisms can significantly improve ransomware detection capabilities. The system provides a scalable and practical framework for protecting modern computing systems against evolving ransomware threats.

5.1 Future Scope

Although the proposed system demonstrates strong performance in ransomware detection, several enhancements can be explored in future work to further improve the system's capabilities.

- **Integration of Additional Behavioral Features:** Future work can incorporate more advanced behavioral indicators such as network traffic patterns, process tree relationships, and API call sequences to improve detection accuracy.
- **Real-Time Deployment in Enterprise Environments:** The system can be extended for large-scale deployment in enterprise networks, where multiple endpoints generate large volumes of logs that must be processed in real time.
- **Online Learning and Adaptive Models:** Machine learning models can be continuously updated using online learning techniques so that the system adapts automatically to newly emerging ransomware variants.
- **Integration with Automated Incident Response Systems:** Future implementations may integrate automated response mechanisms such as process isolation, file recovery, and system rollback to mitigate ransomware attacks immediately after detection.
- **Advanced Deep Learning Architectures:** More sophisticated deep learning models such as Transformers or Graph Neural Networks (GNNs) can be explored to analyze complex behavioral relationships in system events.
- **Cross-Platform Malware Detection:** The framework can be extended to support detection across different operating systems and cloud environments, enabling a

unified ransomware defense system.

In summary, the proposed approach establishes a strong foundation for behavior-based ransomware detection. With further improvements and real-world deployment, such systems can play a significant role in protecting individuals, organizations, and critical infrastructure from modern ransomware attacks.

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Appendix A: Presentation

RANSOMWARE BEHAVIOUR DETECTION MODEL

100% CODE PRESENTATION

Project Guide:
Dr. Jomina John
Associate Professor
Computer Science Department
RSET

Team Members:
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Golda John (U2203102)
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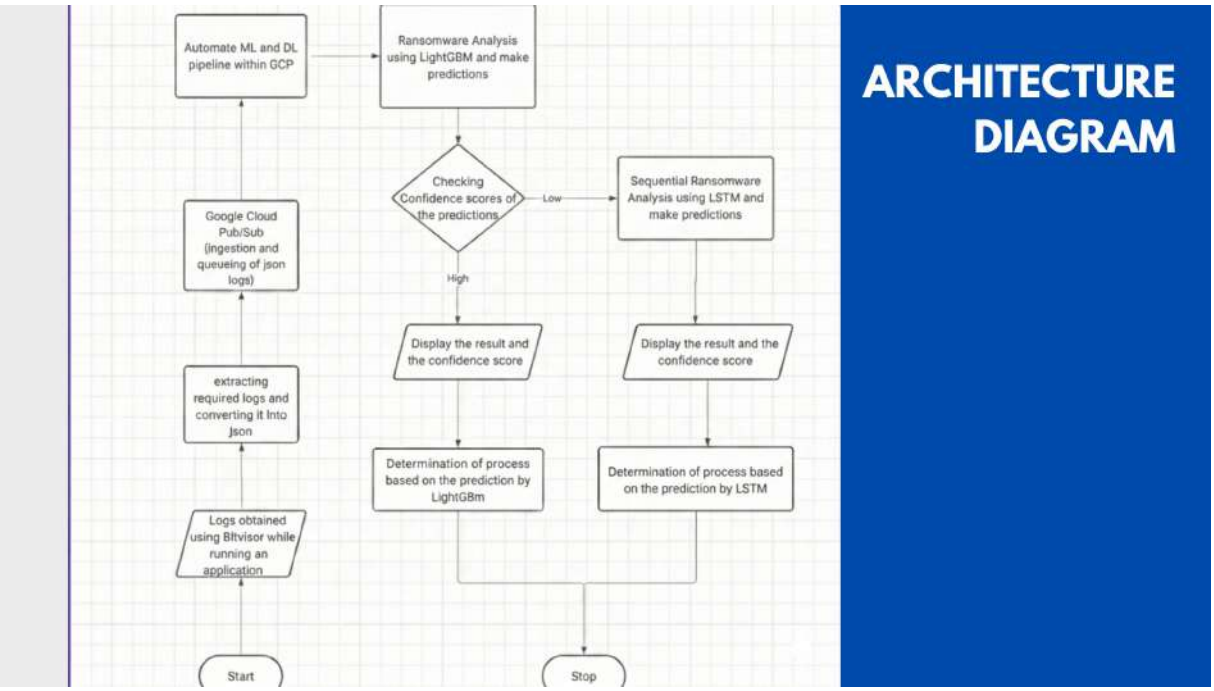
- Problem Definition
- Project Objective
- Architecture Diagram
- Sequence Diagram
- Modules
- Work Breakdown
- Gantt Chart
- Expected Output
- Result
- Conclusion
- References

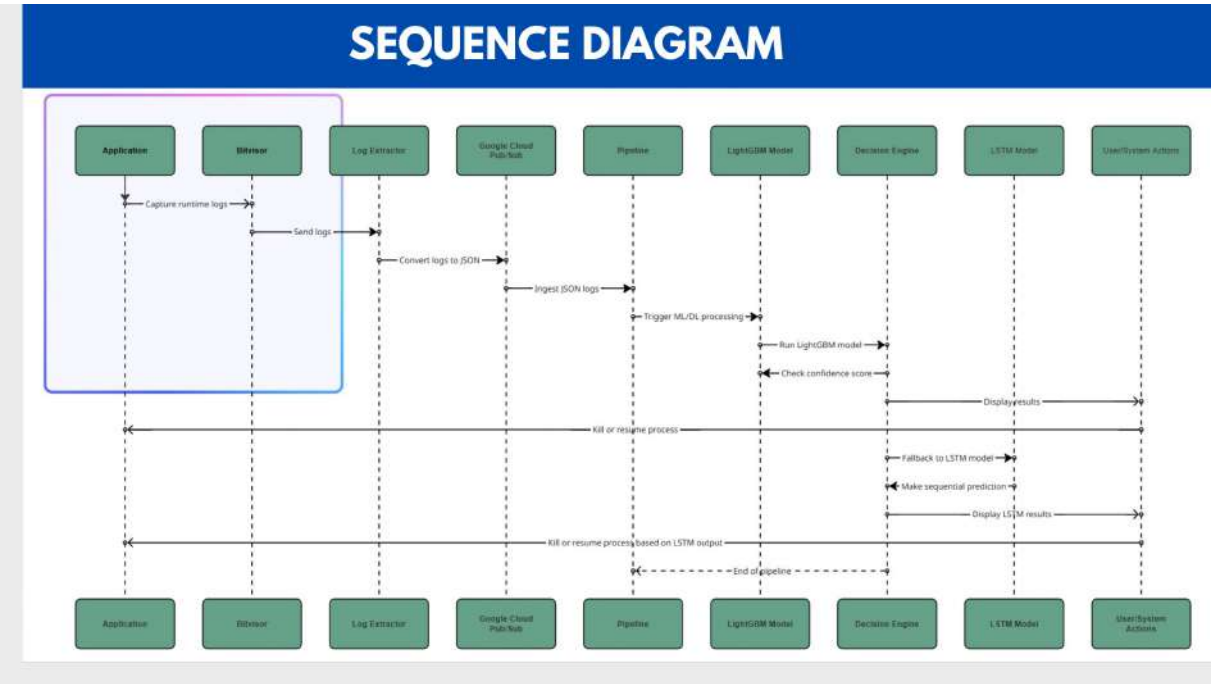
PROBLEM DEFINITION

To create a robust, real-time, behavior-based ransomware detection system that can be effectively used to identify, analyze, and stop both new and evolving ransomware strains at the earliest possible stage of execution—before they are able to encrypt files or inflict any significant damage—by leveraging advanced machine learning techniques such as LightGBM for high-performance classification and LSTM for analyzing sequential behavior patterns over time.

PROJECT OBJECTIVE

- To develop a machine learning-based system to detect ransomware using behavioral data.
- To integrate LightGBM for quick static classification and LSTM for analyzing sequential patterns.
- To deploy models on a cloud platform for scalable, multi-endpoint monitoring. Generate instant alerts to prevent data encryption and system damage.


ARCHITECTURE DIAGRAM
Automate ML and DL pipeline within GCP
Ransomware Analysis using LightGBM and make predictions
Checking Confidence scores of the predictions
Low
High
Sequential Ransomware Analysis using LSTM and make predictions
Google Cloud Pub/Sub (ingestion and queueing of json logs)
Display the result and the confidence score
Display the result and the confidence score
extracting required logs and converting it into Json
Determination of process based on the prediction by LightGBm
Determination of process based on the prediction by LSTM
Logs obtained using Bitvisor while running an application
Start
Stop


SEQUENCE DIAGRAM
Application
Bitvisor
Log Extractor
Google Cloud Pub/Sub
Pipeline
LightGBM Model
Decision Engine
LSTM Model
User/System Actions
Capture runtime logs
Send logs
Convert logs to JSON
Ingest JSON logs
Trigger ML/DL processing
Run LightGBM model
Check confidence score
Display results
Kill or resume process
Fallback to LSTM model
Make sequential prediction
Display LSTM results
Kill or resume process based on LSTM output
End of pipeline

MODULES

1. **BitVisor (Data Collection):**
   - Captures low-level system logs (file, process, system calls).
   - Ensures secure and reliable data monitoring.

2. **Pipelining & Vertex AI (Data Processing):**
   - Cleans, structures, and routes logs between models.
   - Manages dataset storage, training, and evaluation.

3. **LightGBM (Fast Detection):**
   - Quickly classifies normal vs suspicious activity.
   - Low-confidence results are sent to LSTM.

4. **LSTM (Deep Analysis):**
   - Examines log sequences over time.
   - Detects hidden or evolving ransomware behavior.

BITVISOR

- BitVisor runs beneath the OS and stays idle until certain events occur.
- When an app accesses disk or memory, it will trigger a VM-exit, handing control to BitVisor.
- BitVisor captures the event, reads the CR3 register to identify the process, and extracts details like operation type, disk address (LBA), GPA ,size etc.
- This allows efficient logging of key system activity with minimal overhead.

BITVISOR

- BitVisor is a Type-1 hypervisor that runs directly on hardware and sits below the guest OS.
- It does not log anything by default that matches RanSAP/RanSMAP structure but it provides the foundation to implement the modifications needed to produce RanSAP / RanSMAP-style logs
- It allows you to insert hooks at exactly the right place in the hardware interface path, like storage MMIO.

PIPELINING & VERTEX AI

- This pipeline step focuses on collecting, preprocessing and transferring hypervisor-level logs to the cloud for ransomware detection.
- Hypervisor logs capture low-level virtual machine (VM) activities, making them crucial for detecting ransomware that operates at the virtualization layer.
- The data is securely uploaded to Google Cloud Platform (GCP) and processed via pipeline for automated model training and prediction.
- Ensures a scalable, cloud-native, and intelligent detection system capable of handling large volumes of virtualized environment data.

LIGHTGBM

- Trains a Light Gradient Boosting Machine (LightGBM) model using logs from a customized BitVisor environment.
- Extracted logs include file operations, process activity, disk access patterns, and other behavioural indicators.
- The model learns to distinguish ransomware-like activity from normal system behaviour.
- Ensures fast, interpretable, and scalable classification for real-time detection.
- Acts as the first detection stage before escalation to the LSTM module for sequential analysis.

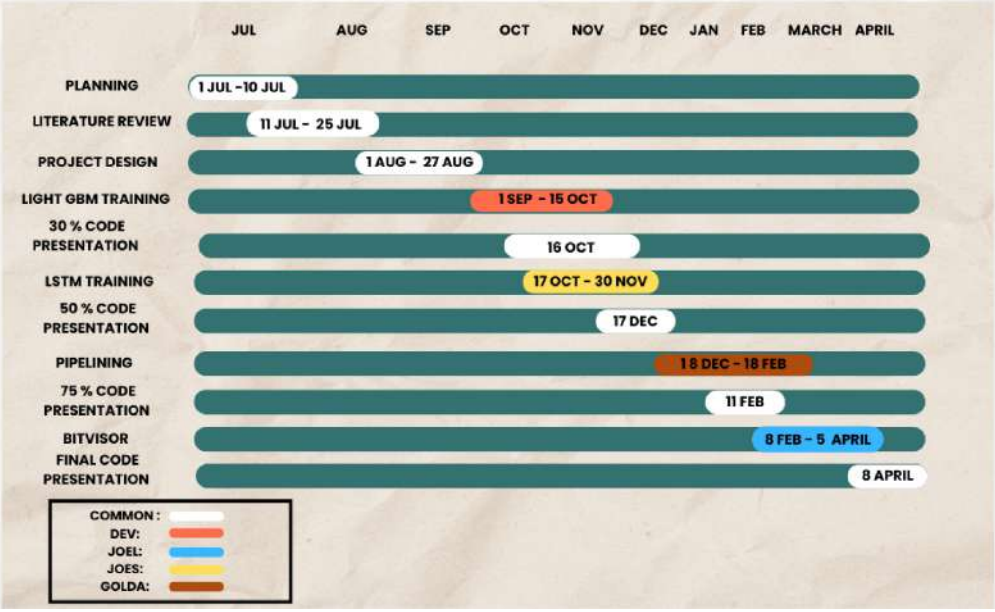
LSTM

- LSTM (Long Short-Term Memory) is a Recurrent Neural Network (RNN) variant used to analyze sequences of low-level system events
- BitVisor logs reflect actual system behavior at the virtualization layer; because these behaviors evolve sequentially as ransomware executes, the LSTM is ideal for capturing temporal dependencies that reveal how malicious behavior unfolds over time.
- Ransomware doesn't act in a single suspicious event; it performs repetitive encrypted I/O operations, rapid file renames, or consistent memory-access spikes.
- These patterns only become clear when observed over time, not per individual event.

WORK BREAKDOWN

Member	Module	Responsibilities
Joel John	Bitvisor simulation and pipelining	Modifying Bitvisor and obtaining the required logs.
Golda John	Pipelining	Data preprocessing, integration and model workflow automation
Dev Sebastian Joseph	LightGBM	Model training, optimization and initial threat classification
Joes Jacob Paul	LSTM	Sequence modeling, temporal analysis and advanced threat detection

GANTT CHART



RESULT

LightGBM

AGGREGATED FEATURES

- count — Total number of write operations
- size_mean — Mean write size
- size_std — Standard deviation of write sizes
- entropy_mean — Mean entropy of written data
- entropy_std — Entropy variability
- entropy_median — Median entropy
- zero_entropy_frac — Fraction of zero-entropy writes
- high_entropy_frac — Fraction of high-entropy writes
- unique_lba_count — Unique logical block addresses accessed
- spike_entropy_count — Sudden entropy transitions

RESULT

LightGBM

```
=== Evaluation (pred_mean) ===
AUC: 1.0000, Average Precision(AP): 1.0000
Classification Report:
              precision    recall  f1-score   support

           0     1.0000    1.0000    1.0000        50
           1     1.0000    1.0000    1.0000        70

    accuracy                         1.0000       120
   macro avg     1.0000    1.0000    1.0000       120
weighted avg     1.0000    1.0000    1.0000       120

Confusion Matrix:
 [[50  0]
 [ 0 70]]
```

RESULT

```
=== Test results ===
AUC = 0.951781
Average Precision (AP) = 0.974429
Classification report:
              precision    recall  f1-score   support

           0     0.8199    0.8664    0.8425       846
           1     0.9150    0.8831    0.8987      1377

    accuracy                         0.8767      2223
   macro avg     0.8674    0.8748    0.8706      2223
weighted avg     0.8788    0.8767    0.8774      2223

Confusion matrix:
 [[ 733  113]
 [ 161 1216]]
Saved metrics to metrics.json

Done.
```

RESULT

Combined Dataset

_s	time_ns	GPA	size	entropy	type	label	source	file_id	LBA	ignore	timestamp	t_rel
1684893977	194817096	18618559496	4096	0.8198624	2	1	mem	AESCrypt-20230524_20-18-05			1.68E+18	
1684893977	198389704	4276093104	4096	0.17081882	2	1	mem	AESCrypt-20230524_20-18-05			1.68E+18	357
1684893977	198910352	6470762232	4096	0.52377657	2	1	mem	AESCrypt-20230524_20-18-05			1.68E+18	409
1684893977	199257092	6474102816	4096	0.7013603	2	1	mem	AESCrypt-20230524_20-18-05			1.68E+18	444
1684893977	200457265	19204565860	4096	0.83580908	2	1	mem	AESCrypt-20230524_20-18-05			1.68E+18	564
1684893977	201031588	19206200004	4096	0.8280477	2	1	mem	AESCrypt-20230524_20-18-05			1.68E+18	621
1684893977	201531839	6532642116	4096	0.83645875	2	1	mem	AESCrypt-20230524_20-18-05			1.68E+18	671
1684893977	202221026	6546334116	4096	0.82959337	2	1	mem	AESCrypt-20230524_20-18-05			1.68E+18	740
1684893977	203428713	19286237252	4096	0.82238922	2	1	mem	AESCrypt-20230524_20-18-05			1.68E+18	861
1684893977	203683133	6660998404	4096	0.77123849	2	1	mem	AESCrypt-20230524_20-18-05			1.68E+18	886
1684893977	204262823	6819719780	4096	0.80338049	2	1	mem	AESCrypt-20230524_20-18-05			1.68E+18	944
1684893977	205193547	6549570020	4096	0.84563087	2	1	mem	AESCrypt-20230524_20-18-05			1.68E+18	1037
1684893977	205707754	7612710904	4096	0.69644233	2	1	mem	AESCrypt-20230524_20-18-05			1.68E+18	1089
1684893977	206065229	18906502420	4096	0.75050312	2	1	mem	AESCrypt-20230524_20-18-05			1.68E+18	1124
1684893977	206305693	18883691316	4096	0.84740983	2	1	mem	AESCrypt-20230524_20-18-05			1.68E+18	1148
1684893977	206754416	7905269844	4096	0.80410868	2	1	mem	AESCrypt-20230524_20-18-05			1.68E+18	1193
1684893977	210730660	18413655544	4096	0.76351475	2	1	mem	AESCrypt-20230524_20-18-05			1.68E+18	1591
1684893977	227700548	18539459076	4096	0.69049538	2	1	mem	AESCrypt-20230524_20-18-05			1.68E+18	3288

RESULT

Memory features

- mem_write_count
- mem_avg_entropy
- mem_std_entropy
- mem_total_bytes
- mem_gpa_variance

Disk features

- disk_write_count
- disk_avg_entropy
- disk_total_bytes
- disk_lba_variance

RESULT

Sequence Files Generated

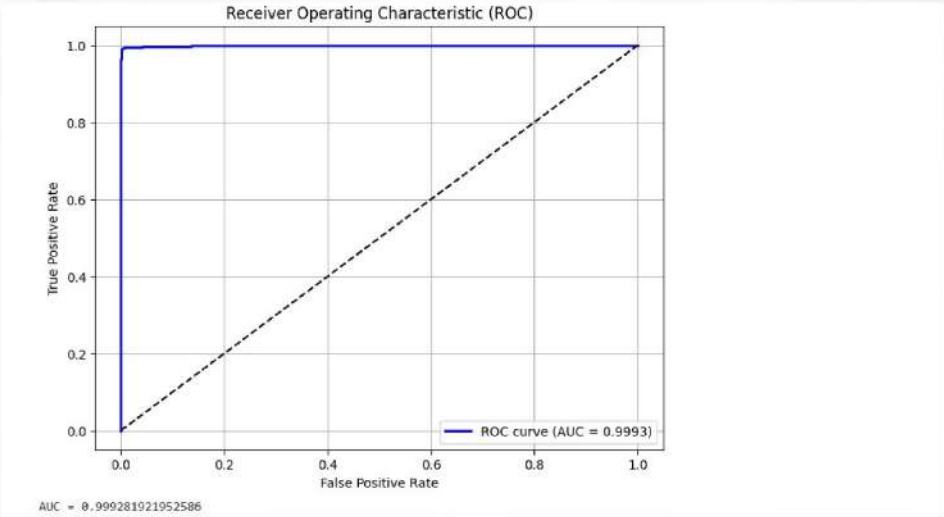
- X_sequences.npy -> shape (150,877, 50, 10)
- y_labels.npy -> shape (150,877,)
- run_map.npy -> mapping each sequence → source run name

RESULT

LSTM

```
===== TEST PERFORMANCE =====
808/808 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step
              precision    recall  f1-score   support

           0       0.98      1.00      0.99      6864
           1       1.00      0.99      1.00     18970

    accuracy                           0.99     25834
   macro avg       0.99      0.99      0.99     25834
weighted avg       0.99      0.99      0.99     25834

Confusion Matrix:
[[ 6834    30]
 [  146 18824]]
```

RESULT



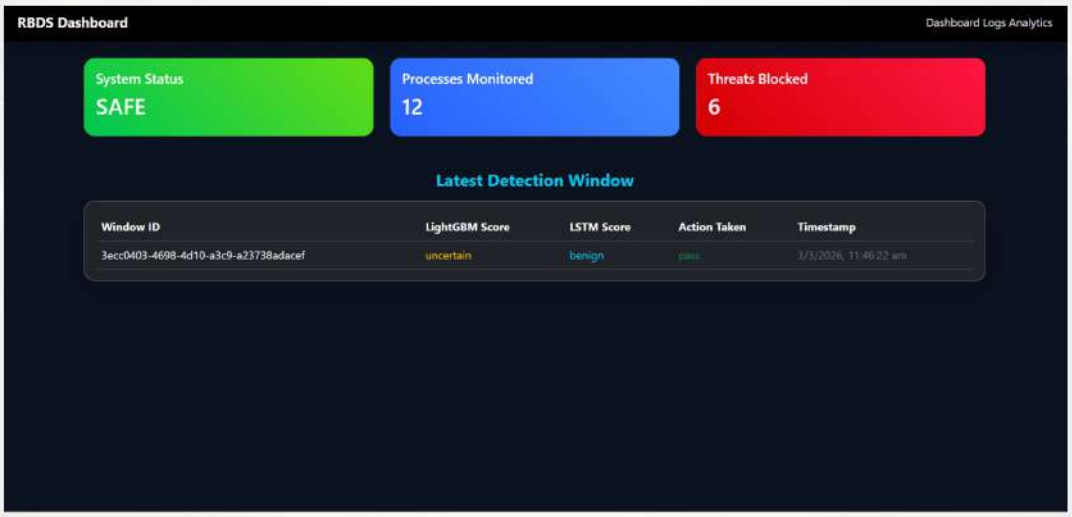
RESULT

```
==============================
📊 DARKSIDE SEQUENCE RESULTS
==============================
Total sequences      : 1373
Ransomware sequences : 1356
Benign sequences     : 17
Ransomware %         : 98.76%
Benign %             : 1.24%

==============================
🧠 DARKSIDE RUN DECISION
==============================
Average probability : 0.9994
FINAL DECISION      : RANSOMWARE
```

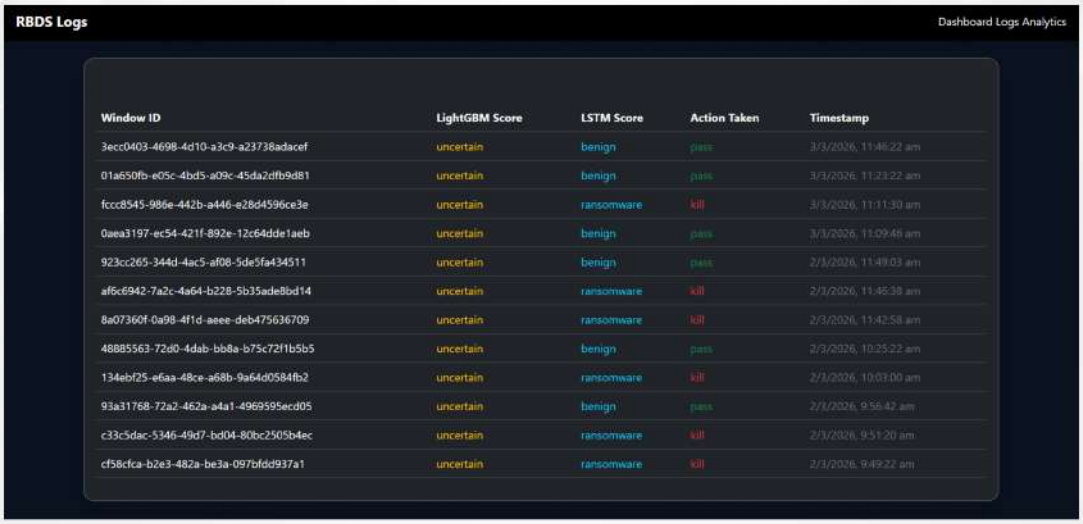
RESULT

```
==============================
📊 FIREFOX SEQUENCE RESULTS
==============================
Total sequences      : 987
Ransomware sequences : 17
Benign sequences     : 970
Ransomware %         : 1.72%
Benign %             : 98.28%

==============================
🧠 FIREFOX RUN DECISION
==============================
Average probability : 0.0426
FINAL DECISION      : BENIGN
```

RESULT



RESULT



Window ID	LightGBM Score	LSTM Score	Action Taken	Timestamp
3ecc0403-4698-4d10-a3c9-a23738adacef	uncertain	benign	pass	3/3/2026, 11:46:22 am
01a650fb-e05c-4bd5-a09c-45da2dfb9d81	uncertain	benign	pass	3/3/2026, 11:23:22 am
fccc8545-986e-442b-a446-e28d4596ce3e	uncertain	ransomware	kill	3/3/2026, 11:11:30 am
0aea3197-ec54-421f-892e-12c64dde1aeb	uncertain	benign	pass	3/3/2026, 11:09:48 am
923cc265-344d-4ac5-af08-5de5fa434511	uncertain	benign	pass	2/3/2026, 11:49:03 am
af6c6942-7a2c-4a64-b228-5b35ade8bd14	uncertain	ransomware	kill	2/3/2026, 11:46:38 am
8a07360f-0a98-4f1d-aeee-deb475636709	uncertain	ransomware	kill	2/3/2026, 11:42:58 am
48885563-72d0-4dab-bb8a-b75c72f1b5b5	uncertain	benign	pass	2/3/2026, 10:25:22 am
134ebf25-e6aa-48ce-a68b-9a64d0584fb2	uncertain	ransomware	kill	2/3/2026, 10:03:00 am
93a31768-72a2-462a-a4a1-4969595ecd05	uncertain	benign	pass	2/3/2026, 9:56:42 am
c33c5dac-5346-49d7-bd04-80bc2505b4ec	uncertain	ransomware	kill	2/3/2026, 9:51:20 am
cf58cfca-b2e3-482a-be3a-097bfdd937a1	uncertain	ransomware	kill	2/3/2026, 9:49:22 am

RESULT

CONCLUSION

- Ransomware causes severe damage once encryption begins.
- Our approach focuses on early detection through behavior monitoring.
- LightGBM enables fast, real-time anomaly checks.
- LSTM captures deeper behavioral patterns over time.
- Together, these methods create a practical and reliable defense.
- The goal is to protect users and organizations from one of today's most dangerous cyber threats.

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THANK YOU

Appendix B:
Vision, Mission, Programme Outcomes and Course Outcomes

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Vision, Mission, Programme Outcomes and Course Outcomes

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Programme Outcomes (PO)

Engineering Graduates will be able to:

PO1: Engineering Knowledge

Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis

Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

PO3: Design/Development of Solutions

Design creative solutions for complex engineering problems and design/develop systems/-components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

PO4: Conduct Investigations of Complex Problems

Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.

PO5: Engineering Tool Usage

Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.

PO6: The Engineer and The World

Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

PO7: Ethics

Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.

PO8: Individual and Collaborative Team work

Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication

Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

PO10: Project Management and Finance

Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning

Recognize the need for, and have the preparation and ability for i) independent and

life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

Programme Specific Outcomes (PSO)

A graduate of the Computer Science and Engineering Program will demonstrate:

PSO1: Computer Science Specific Skills
The ability to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas by understanding the core principles and concepts of computer science and thereby engage in national grand challenges.

PSO2: Programming and Software Development Skills
The ability to acquire programming efficiency by designing algorithms and applying standard practices in software project development to deliver quality software products meeting the demands of the industry.

PSO3: Professional Skills
The ability to apply the fundamentals of computer science in competitive research and to develop innovative products to meet the societal needs thereby evolving as an eminent researcher and entrepreneur.

Course Outcomes (CO)

After the completion of the course the student will be able to:

CO1: Model and solve real-world problems by applying knowledge across domains (Cognitive knowledge level: Apply).

CO2: Develop products, processes or technologies for sustainable and socially relevant applications (Cognitive knowledge level: Apply).

CO3: Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks (Cognitive knowledge level: Apply).

CO4: Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms (Cognitive knowledge level: Apply).

CO5: Identify technology/research gaps and propose innovative/creative solutions

(Cognitive knowledge level: Analyze).

CO6: Organize and communicate technical and scientific findings effectively in written and oral forms (Cognitive knowledge level: Apply).

Appendix C: CO-PO-PSO Mapping

COURSE OUTCOMES:

SNO	DESCRIPTION	BLOOM'S TAXONOMY LEVEL
1	Model and solve real world problems by applying knowledge across domains.	Apply
2	Develop products, processes or technologies for sustainable and socially relevant application.	Apply
3	Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks.	Apply
4	Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms.	Apply
5	Identify technology/research gaps and propose innovative/creative solutions.	Analyze
6	Organize and communicate technical and scientific findings effectively in written and oral forms.	Apply

MAPPING COURSE OUTCOMES (COs) – PROGRAM OUTCOMES (POs) AND COURSE OUTCOMES (COs) – PROGRAM SPECIFIC OUTCOMES (PSOs)

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
1	2	2	2	1		2	1	1	1	1	1	2	2	2
2	2	2	2		2	3	2	1	1	1	1	2	2	
3								3	2	2	1			3
4		1	2		3	2	3	2	2	3	1		2	2
5	2	3	3	1	2					1	3	3		
6			2			2	2	2	3	1	1			3

JUSTIFICATIONS FOR CO-PO MAPPING

Mapping	LOW/ MEDIUM /HIGH	Justification
101003/CS822U.CO1-PO1	Medium	Applies engineering knowledge, mathematics, and computing fundamentals to model and solve real-world problems.
101003/CS822U.CO1-PO2	Medium	Requires identification, formulation and analysis of engineering problems before developing solutions.
101003/CS822U.CO1-PO3	Medium	Involves designing systems or solutions to address identified engineering needs.
101003/CS822U.CO1-PO4	Low	Limited investigation or experimentation may be required to validate the developed models or solutions.
101003/CS822U.CO1-PO6	Medium	Solutions developed may influence societal, environmental, or sustainability aspects of engineering applications.
101003/CS822U.CO1-PO7	Medium	Encourages consideration of ethical and professional responsibilities while developing solutions.
101003/CS822U.CO1-PO8	Low	Some level of teamwork may be required while working on modeling or project tasks.
101003/CS822U.CO1-PO9	Low	Requires communication of results and findings through reports or presentations.
101003/CS822U.CO1-PO10	Low	Involves basic planning and execution of tasks while implementing solutions.
101003/CS822U.CO1-PO11	Low	Encourages independent learning and adapting new knowledge to solve real-world problems.
101003/CS822U.CO1-PSO1	Medium	Uses core computer science principles to analyze problems and design appropriate technical solutions.
101003/CS822U.CO1-PSO2	Medium	Involves algorithm design and application of software development practices to implement solutions.
101003/CS822U.CO1-PSO3	Medium	Applies computer science fundamentals in research or innovative problem solving for societal needs.
101003/CS822U.CO2-PO1	Medium	Utilizes engineering knowledge to design and develop products or systems.

Mapping	LOW/ MEDIUM /HIGH	Justification
101003/CS822U.CO2-PO2	Medium	Requires analysis of user needs and technological feasibility before development.
101003/CS822U.CO2-PO3	Medium	Focuses on design and development of solutions addressing real-world requirements.
101003/CS822U.CO2-PO5	Medium	Requires use of modern engineering and IT tools for system or product development.
101003/CS822U.CO2-PO6	High	Emphasizes sustainability and societal impact during solution development.
101003/CS822U.CO2-PO7	Medium	Ethical and professional responsibilities are considered when developing socially relevant technologies.
101003/CS822U.CO2-PO8	Low	Development activities may require collaboration among team members.
101003/CS822U.CO2-PO9	Low	Involves communicating design ideas and development outcomes.
101003/CS822U.CO2-PO10	Low	Requires basic planning and coordination during development tasks.
101003/CS822U.CO2-PO11	Low	Encourages continuous learning of emerging technologies while building solutions.
101003/CS822U.CO2-PSO1	Medium	Applies computer science fundamentals to design technically feasible solutions.
101003/CS822U.CO2-PSO2	Medium	Uses programming and software development practices to implement solutions.
101003/CS822U.CO3-PO8	High	Strongly relates to functioning effectively as a member or leader in multidisciplinary teams.
101003/CS822U.CO3-PO9	Medium	Requires effective communication among team members during collaboration.
101003/CS822U.CO3-PO10	Medium	Team leadership involves project coordination and task management.
101003/CS822U.CO3-PO11	Low	Team interaction supports learning from peers and adapting new approaches.
101003/CS822U.CO3-PSO3	High	Professional collaboration and teamwork support innovative computer science solutions.
101003/CS822U.CO4-PO2	Low	Requires analysis and planning before executing project tasks.

Mapping	LOW/ MEDIUM /HIGH	Justification
101003/CS822U.CO4-PO3	Medium	Involves designing and implementing project solutions based on planned activities.
101003/CS822U.CO4-PO5	High	Requires effective use of modern tools and resources during project execution.
101003/CS822U.CO4-PO6	Medium	Considers societal and environmental impact while implementing solutions.
101003/CS822U.CO4-PO7	High	Emphasizes adherence to ethical practices and professional responsibilities.
101003/CS822U.CO4-PO8	Medium	Task execution may involve teamwork and coordination among team members.
101003/CS822U.CO4-PO9	Medium	Requires communication of progress, outcomes and documentation.
101003/CS822U.CO4-PO10	High	Focuses on project management including scheduling, resource allocation and task execution.
101003/CS822U.CO4-PO11	Low	Encourages adaptability and continuous learning during project implementation.
101003/CS822U.CO4-PSO2	Medium	Uses software development practices to execute and manage project tasks.
101003/CS822U.CO4-PSO3	Medium	Applies computer science knowledge in practical project development environments.
101003/CS822U.CO5-PO1	Medium	Uses engineering knowledge to understand technological limitations and opportunities.
101003/CS822U.CO5-PO2	High	Requires deep problem analysis and identification of research gaps.
101003/CS822U.CO5-PO3	High	Proposes innovative design solutions to address identified gaps.
101003/CS822U.CO5-PO4	Low	May involve limited investigation or validation of proposed ideas.
101003/CS822U.CO5-PO5	Medium	Uses appropriate tools and techniques to analyze and develop innovative solutions.
101003/CS822U.CO5-PO10	Low	Requires minimal planning while proposing new solution approaches.
101003/CS822U.CO5-PO11	High	Strongly promotes independent learning and exploration of emerging technologies.
101003/CS822U.CO5-PSO1	High	Applies computer science fundamentals to identify and solve complex technical problems.

Mapping	**LOW/ MEDIUM /HIGH**	**Justification**
101003/CS822U.CO6-PO3	Medium	Requires presentation of design concepts and developed solutions.
101003/CS822U.CO6-PO6	Medium	Communication may include discussion of societal or environmental implications.
101003/CS822U.CO6-PO7	Medium	Requires responsible and ethical communication of technical results.
101003/CS822U.CO6-PO8	Medium	Communication supports collaboration within team environments.
101003/CS822U.CO6-PO9	High	Strongly relates to effective written and oral communication of technical findings.
101003/CS822U.CO6-PO10	Low	Requires structured reporting and documentation in project management contexts.
101003/CS822U.CO6-PO11	Low	Encourages continuous improvement in communication and presentation skills.
101003/CS822U.CO6-PSO3	High	Communication of innovative computer science solutions and research outcomes.